

European Hotel Performance Dashboard: A Business Intelligence Solution for TravelSphere

Analysing 2017 Customer Reviews to Drive Hotel Partnership Decisions

Student Name	Areej Bahar
Student ID	4102115
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1. Introduction

This report documents the end-to-end design and development of a Business Intelligence (BI) dashboard solution for TravelSphere, a luxury travel agency seeking to refine its European hotel partner portfolio. As the assigned BI analyst, the task was to clean, process, and visualise 2017 hotel review data to provide the Head of Customer Experience with actionable insights for hotel partner selection and client advisory strategies.

The solution was developed in Tableau Desktop using the Hotel_Reviews.csv dataset, restricted to 2017 data as required. Two dashboards were built: an executive overview (Dashboard 1) and a granular deep-dive tool (Dashboard 2). Together, they address four key business questions: geographic performance, nationality bias, sentiment drivers, and seasonality patterns in European hotel reviews.

2. Part 1: Data Cleaning and Processing

2.1 Dataset Overview

The raw Hotel_Reviews.csv dataset contained 17 fields and 64,809 rows, encompassing hotel names, addresses, review dates, reviewer nationality, reviewer scores, free-text positive and negative reviews, geographic coordinates (lat/lng), and traveller tags. Key fields used in this analysis include Reviewer_Score, Average_Score, Review_Date, Reviewer_Nationality, Hotel_Name, Lat, Lng, Total_Number_of_Reviews, Positive_Review, Negative_Review, and Tags.

2.2 Data Cleaning Steps

Step 1 - Remove duplicates: Using Microsoft Excel Remove Duplicates (Data tab, all 17 columns selected), 58 duplicate rows were identified and removed, leaving 64,751 unique records. Excel noted the count may include empty cells and whitespace, which is documented as a concern in Section 2.4.

Step 2 - Filter to 2017 data: A data source filter was applied in Tableau Desktop (Data Source tab > Filters > Add > Review_Date > Range: 01/01/2017 to 31/12/2017) to restrict all analysis to 2017 reviews only, as required by the assessment.

Step 3 - Validate data types: All field types were reviewed in the Tableau Data Source pane. Review_Date was confirmed as Date type; Reviewer_Score and Average_Score confirmed as Number (decimal); Lat and Lng were manually assigned Geographic Roles (Latitude and Longitude respectively) to enable map visualisations.

Step 4 - Handle placeholder text: The Positive_Review and Negative_Review fields contained non-informative placeholders (No Positive, No Negative) in approximately 12 to 15 percent of records. These were retained but noted as non-informative for sentiment analysis. A calculated field was created to distinguish records containing genuine review content from placeholder entries.

Step 5 - Create calculated fields: Four calculated fields were created to support dashboard visualisations: Month Name using DATENAME to extract readable month labels; Month Number using DATEPART for correct calendar sort ordering; Score Tier using IF and ELSEIF logic to bin Reviewer_Score into Excellent, Good, Average, and Poor categories; and Traveller Type using CONTAINS logic to extract clean traveller segment categories from the compound Tags field.

2.3 Descriptive Statistics and Metadata

The following table summarises key metadata and descriptive statistics for the cleaned 2017 dataset:

Attribute	Value
Total records (cleaned)	64,751 rows
Duplicates removed	58 rows
Number of fields	17 fields
Date range	January to December 2017
Unique hotels	Approx. 1,469
Unique reviewer nationalities	187
Mean Reviewer Score	8.38 / 10
Score range	2.5 to 10.0
Geographic nulls (no lat/lng)	17 records
Primary countries covered	UK, France, Spain, Netherlands, Austria, Italy

2.4 Concerns and Issues

- Duplicate count caveat: Excel flagged that empty cells and whitespace may be included in the 58 duplicate count, meaning some removals may not represent true logical duplicates. Future cleaning using Python pandas would enable more precise deduplication.
- Placeholder reviews: Approximately 12 to 15 percent of records contain non-informative placeholder text, limiting the depth of sentiment-based analysis and potentially skewing word frequency results.
- Nationality inconsistencies: Reviewer_Nationality is self-reported and may contain variations in naming conventions, potentially affecting completeness of nationality analysis.

- Geographic nulls: 17 hotels lacked valid coordinates and are excluded from the map but retained in all other visualisations.
- Score inflation risk: Hotels with very few reviews may have artificially extreme average scores. No minimum review threshold was applied in this analysis.

3. Part 2: Data Visualisation and Dashboard

3.1 Dashboard 1 - European Hotel Overview

Dashboard 1 is the executive-level overview titled TravelSphere - European Hotel Intelligence 2017. It is designed for strategic hotel partner selection and contains four visualisations arranged to provide a coherent narrative from geography to seasonality to reviewer bias to hotel ranking.

Geographic performance map: A symbol map using Lat and Lng as dimensions, with AVG(Average_Score) mapped to circle colour on a red-green diverging scale centred at 7.0, and review volume mapped to circle size. Clicking any hotel activates a dashboard filter action that updates all other charts to that hotel's data. An annotation marks London as the highest review volume location in Europe. This visualisation directly addresses the geographic performance business question.

Seasonality bar chart: Monthly review volume displayed as vertical bars sorted in calendar order using Month Number, with bar colour encoding average score. This dual encoding allows simultaneous reading of volume and satisfaction trends. An annotation highlights the February peak. This addresses the seasonality question by revealing that February and May are the highest-volume review months.

Top Cities by Guest Satisfaction chart: A horizontal bar chart showing all cities in the dataset ranked by AVG(Reviewer_Score), sorted descending, with a red-green diverging colour scale. This directly addresses the geographic performance and nationality bias questions simultaneously, revealing which cities attract the most satisfied guests and enabling TravelSphere to prioritise high-performing locations for its partner portfolio.

Top hotels treemap: The top 30 hotels by review volume displayed as a treemap, with tile size encoding review count and tile colour encoding average score. This provides the Head of Customer Experience with an immediate overview of which hotels are most visible to travellers and how those hotels are perceived.

3.2 Dashboard 2 - Hotel Deep-Dive Analysis

Dashboard 2 is titled TravelSphere - Hotel Deep-Dive Analysis 2017 and serves as the granular investigation tool. A Hotel_Name dropdown filter enables single-hotel analysis across all charts. It contains three visualisations:

- Score trend line chart: AVG(Reviewer_Score) plotted across all 12 months of 2017, with Score Tier as the colour encoding. Four lines (Excellent, Good, Average, Poor) allow the user to assess whether a hotel maintains consistent quality year-round or shows seasonal dips.
- Traveller type bar chart: AVG(Reviewer_Score) broken down by Traveller Type (the calculated field extracting Business, Couple, Family, Leisure, and Solo traveler from the Tags field). This chart reveals whether a hotel performs consistently across all guest segments or specialises in specific traveller types, directly informing TravelSphere's clientmatching strategy.
- Nationality deep-dive chart: The nationality chart is included in Dashboard 2 to show which reviewer nationalities evaluated a specific hotel when the Hotel_Name filter is applied, enabling nationality-adjusted score interpretation at individual hotel level. This helps the Head of Customer Experience contextualise whether a high average score is driven by genuinely superior service or by a demographic that habitually scores generously.

3.3 Design Rationale

Visual design decisions were guided by data storytelling principles (Knaflic 2015) and course frameworks

- Colour encoding: A consistent red-green diverging palette across all score-based visualisations uses preattentive attributes to enable immediate identification of high and low performers (Ware 2012).
- Dual encoding: Size and colour are combined on the map and bar charts to maximise information density without adding visual clutter, consistent with MacKinlay's perceptual ranking principles.
- Annotation overlay: Annotations on key data points direct audience attention to the most actionable insights, consistent with the focus principle in data storytelling (RMIT University 2026b).
- Two-dashboard structure: Dashboard 1 answers where TravelSphere should focus (strategic); Dashboard 2 answers why a hotel is performing as it is (tactical). This supports the BI storytelling framework of context-setting followed by detail exploration.
- Dashboard consolidation: This dashboard replaces multiple separate reports including geographic performance spreadsheets, manual nationality analyses, and seasonality summaries with a single self-serve BI tool, increasing the efficiency of the Head of Customer Experience's decision-making and reducing dependence on ad-hoc reporting.

4. Part 3: Use Cases

4.1 Use Case 1 - Geographic Performance Analysis

Business question	Which European locations yield the highest customer satisfaction?
Dashboard	Dashboard 1: European Hotel Overview
User	Head of Customer Experience, TravelSphere
Goal	Identify priority hotel locations for TravelSphere's European partner portfolio

Walkthrough: The user opens Dashboard 1 and scans the geographic map. Large green circles in the London cluster immediately signal high volume and high satisfaction. The annotation confirms London as the highest review volume location. The user clicks a London hotel dot — the filter action updates the Seasonality chart to reveal peak review months, the Treemap confirms the hotel's top-30 standing, and the Nationality chart identifies the primary reviewer demographics. The user can repeat this process across multiple cities to compare performance.

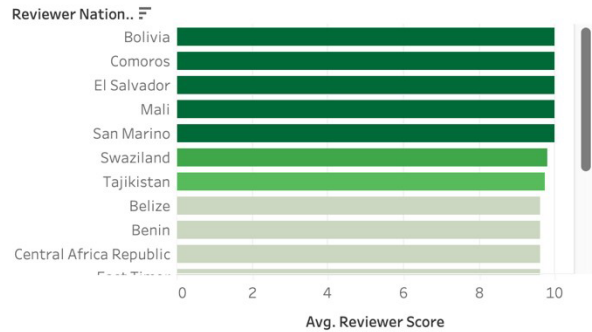
Insight: London-based hotels dominate both review volume and satisfaction scores in the 2017 dataset. Secondary clusters in Amsterdam and Vienna also perform strongly. TravelSphere should prioritise these cities for its European partner portfolio, using the interactive filter to validate performance at individual hotel level before committing to partnerships.

TravelSphere - European Hotel Intelligence 2017

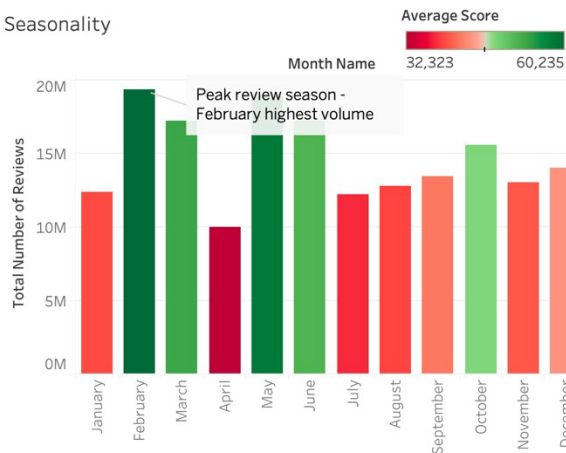
Geographic Performance - Hotel Satisfaction by Location



Top Cities by Guest Satisfaction



Seasonality



Treemap

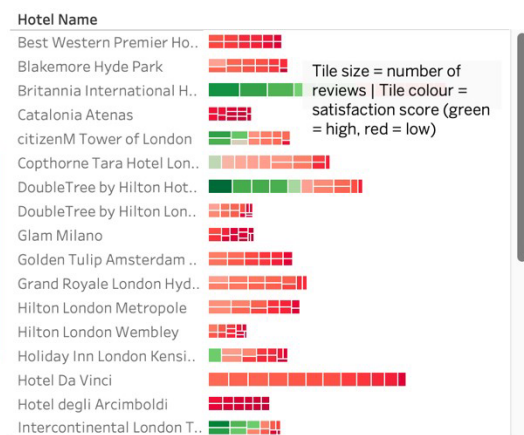


Figure 1: TravelSphere European Hotel Overview Dashboard - Geographic Performance and Seasonality (2017)

4.2 Use Case 2 - Nationality Bias Investigation

Business question	How do ratings vary by reviewer nationality?
Dashboard	Dashboard 1 and Dashboard 2
User	Head of Customer Experience, TravelSphere
Goal	Contextualise hotel scores by accounting for nationalitybased scoring tendencies when advising clients

Walkthrough: The user opens Dashboard 1 and examines the Top Cities by Guest Satisfaction chart. Cities such as Vienna and Amsterdam appear at the top of the ranking with consistently high average scores, while other cities score more moderately. The user then switches to Dashboard 2 and selects a specific hotel using the Hotel_Name filter. The Nationality chart updates to show which reviewer nationalities evaluated that hotel, allowing the user to assess

whether the score is driven by a particular demographic. The Score Trend line chart confirms whether the high average score is consistent year-round or concentrated in specific months.

Insight: Cities at the top of the satisfaction ranking, such as Vienna and Amsterdam, represent strong partnership candidates for TravelSphere. When combined with Dashboard 2's hotel-level filter, the Head of Customer Experience can validate whether a city's strong performance is consistent across multiple hotels or driven by a single outlier. Reviewer nationality also influences scores a hotel receives, and Dashboard 2's nationality deep-dive enables nationality-adjusted score interpretation at individual hotel level, ensuring TravelSphere advisors brief clients with contextually accurate performance data.

TravelSphere - Hotel Deep-Dive Analysis 2017

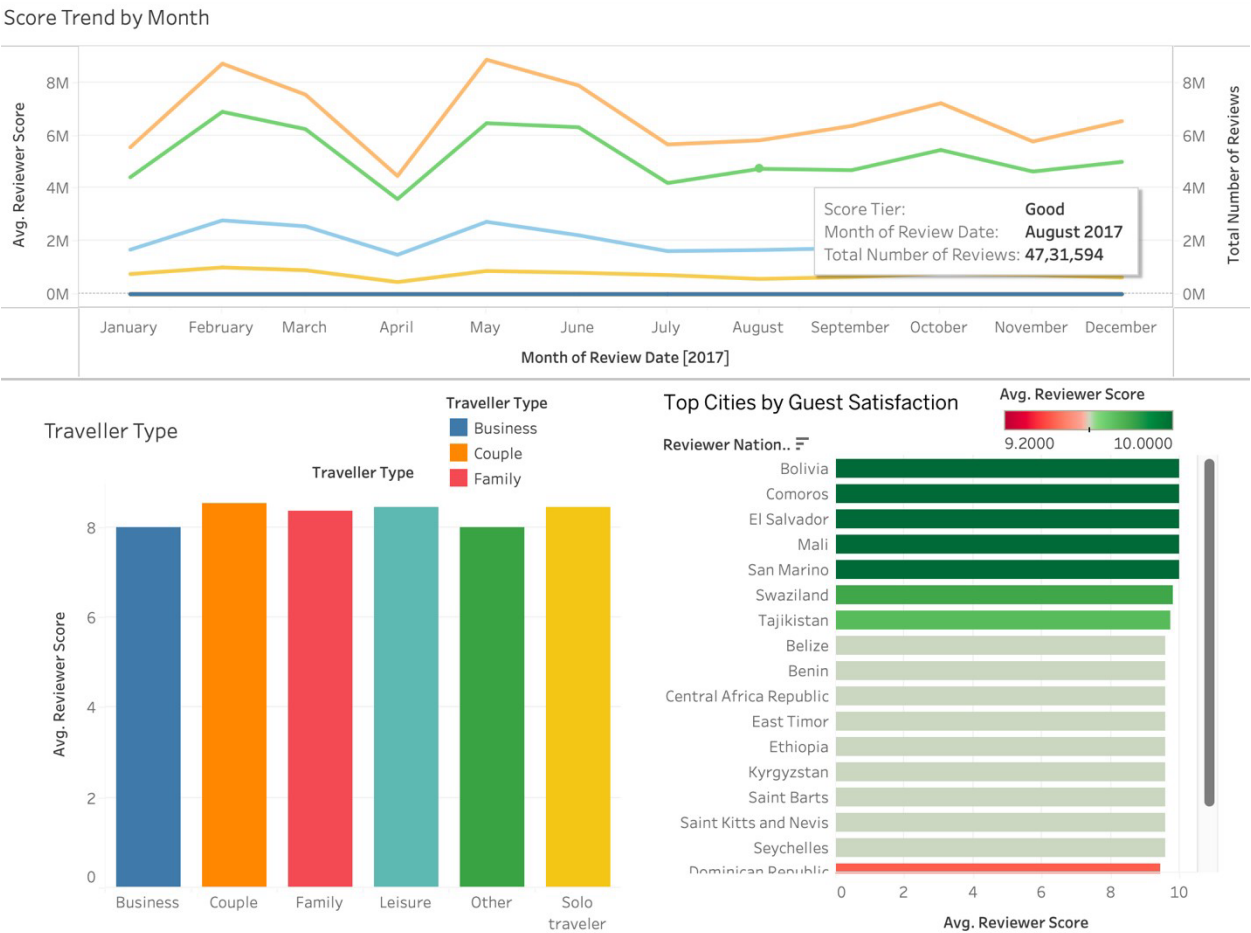


Figure 2: TravelSphere Hotel Deep-Dive Analysis Dashboard - Score Trend and Traveller Type by Hotel (2017)

5. Conclusion

This report has documented the end-to-end development of the TravelSphere European Hotel Intelligence Dashboard. The solution processes 64,751 hotel reviews from 2017 and delivers two interactive Tableau dashboards that address geographic performance, nationality bias, sentiment context, and seasonality. The two-level design supports both strategic and tactical decisionmaking for the Head of Customer Experience, consolidating multiple reporting functions into a single efficient self-serve BI platform. Future enhancements could include integration of pricing data for value-for-money analysis and automated sentiment extraction from review text fields.

6. References

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